

Assignment 2: Develop any distributed application using CORBA to demonstrate object brokering. (Calculator or String operations).

Steps to Perform CORBA Program on Ubuntu

Step 1: Open Ubuntu Terminal

Press:

Ctrl + Alt + T

Step 2: Install Java 8

Update Ubuntu:

```
sudo apt update
```

Install Java 8:

```
sudo apt install openjdk-8-jdk -y
```

Step 3: Check Java Version

```
java -version
```

Step 4: Create Project Folder

```
mkdir CORBA
```

```
cd CORBA
```

Step 5: Create IDL File

```
nano Calc.idl
```

Save the file using:

Ctrl + O → Enter → Ctrl + X

Step 6: Generate CORBA Files

```
idlj -fall Calc.idl
```

This command automatically generates CORBA helper files.

Step 7: Create Server File

```
nano Server.java
```

Paste the server code and save the file.

Step 8: Create Client File

```
nano Client.java
```

Paste the client code and save the file.

Step 9: Compile All Java Files

```
javac *.java
```

Step 10: Start CORBA Naming Service

Open a new terminal and run:

cd CORBA

```
orbd -ORBInitialPort 1050&
```

Step 11: Run Server Program

Open another terminal and run:

cd CORBA

```
java Server -ORBInitialPort 1050
```

Output:

Server Started

Step 12: Run Client Program

Open another terminal and run:

cd CORBA

```
java Client -ORBInitialPort 1050
```

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Viva Questions and Answers for CORBA Assignment

1. What is CORBA?

CORBA stands for:

Common Object Request Broker Architecture

It is used for communication between distributed applications.

2. What is the purpose of CORBA?

CORBA allows objects written in different languages and running on different machines to communicate with each other.

3. What is ORB?

ORB stands for:

Object Request Broker

It acts as middleware between client and server.

4. What is IDL?

IDL means:

Interface Definition Language

It defines methods that client and server can use.

5. Why is Calc.idl file used?

It defines remote methods like:

add(), sub(), mul(), div()

6. What is the use of idlj -fall Calc.idl command?

It generates helper and skeleton files automatically from the IDL file.

7. What is Naming Service in CORBA?

Naming Service helps the client find remote server objects.

8. Which command starts Naming Service?

orbd -ORBInitialPort 1050&

9. Which port is used in CORBA?

Default port used:

1050

10. What is the role of the server?

The server creates remote objects and provides services to clients.

11. What is the role of the client?

The client accesses remote methods from the server.

12. What is distributed application?

An application where client and server run on different systems and communicate over a network.

13. What is middleware?

Middleware is software that connects client and server applications.

Example:

CORBA

14. Why Java 8 is used for CORBA?

Because CORBA support was removed in Java 11 and later versions.

15. What is the function of ORB.init()?

It initializes the CORBA environment.

16. What is rebind() used for?

It registers the server object with the naming service.

17. What is resolve_str() used for?

Client uses it to find the remote object from naming service.

18. What are the advantages of CORBA?

- Platform independent

- Language independent
 - Supports distributed systems
 - Reusable objects
-

19. What are the disadvantages of CORBA?

- Complex architecture
 - Slow performance
 - Difficult debugging
-

20. What is the output of your program?

Addition = 30

Subtraction = 10

Multiplication = 200

Division = 2

21. Real-life applications of CORBA

- Banking systems
 - Telecom systems
 - Enterprise applications
 - Distributed software systems
-

22. Difference between RMI and CORBA

RMI	CORBA
Java only	Language independent
Simple	Complex
Uses Java objects	Uses IDL

23. What happens if server is not started?

Client cannot connect and gives connection error.

24. What files are required in this assignment?

Main files:

Calc.idl

Server.java

Client.java

25. What is object brokering?

Object brokering means communication between distributed objects using ORB.